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Message-based patterns

Distributed Systems Programming

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Overview

We are looking at these topics today:

- Architectural Pattern Types
- Message-based Patterns
- Messaging
- Pipelines
- Transformations

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Patterns

- A design pattern
  - *"… describes a problem which occurs over and over in our environment, and then describes the core of the solution to that problem, in such a way that you can use this solution a million times over, without ever doing it the same way twice."*

[Christopher Alexander – *The Timeless Way of Building - 1977*]

This book proposed a theory of architectural patterns (for buildings but also more generally for any design) which allow replication of quality through shared template solutions to common problems.

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Patterns

Often we use a pattern without realising.

- The best programmers know the patterns already and can choose a selection of patterns that make sense for any given scenario.
- The performance, scalability and stability of our systems are often affected by the pattern(s) we choose, as are the maintainability, readability and expandability.
- It is worth considering that frameworks also use patterns, so by understanding what's going on under the hood we can choose a framework that is right for our purposes.

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Pattern properties

- Context
  - Describes the situation in which the pattern must operate
  - Includes brief descriptions of relevant factors
- Problem
  - Describes the target functionality required
  - Should be as closely defined as possible
- Forces
  - Summary of the external influences on the solution design
  - Lists any pressures, constraints and desirable features of the potential solutions
  - May include conflicting requirements
- Solution
  - A description of the pattern which will solve the described problem
  - Level of detail will depend upon the type of pattern

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"Each pattern then depends both on the smaller patterns it contains, and on the larger patterns within which it is contained."

[Christopher Alexander]

Patterns

- Patterns exist at different levels of abstraction
  - **Architecture patterns**
    - Describes structural organisation of application
    - Predefines required subsystems and their responsibilities
    - Specifies rules and guidelines for organising relationships between subsystems
  - **Design patterns**
    - Suggests how to refine subsystems or components within a system
    - Describes desirable relationships between components
    - Describes a known good solution to a commonly occurring design problem in a particular context
  - **Implementation patterns**
    - Low-level patterns of implementation on a specific platform
    - Describes how to implement designs on a specific system
    - May be language-specific

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Patterns for Distributed Systems

These are the key patterns we'll be concentrating on for distributed systems:

- Message-based patterns
  - Help to integrate the components of a distributed system
  - Specific for distributed systems
- Software design patterns
  - General design patterns
  - Useful for distributed systems
  - Support and complement message-based patterns

Message-based patterns

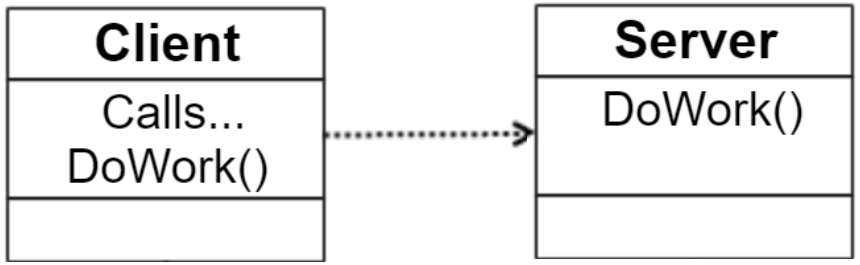
We are going to look at a few basic message-based patterns today

- Point-to-point connections
- Broker
- Message bus

Patterns are EXTREMELY useful in deciding on the core design principles your solution will follow – they are only a pattern though, implementation is still up to you and will be different in different use cases.

I would advise that you continue to research patterns for all software ‘problems’.

Point-to-point pattern

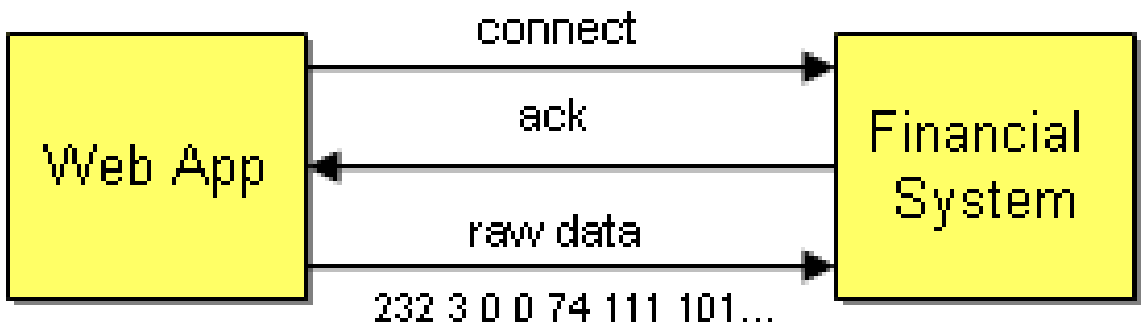


- Simplest pattern
  - One receiver per message
  - No routing
  - Basic client-server

- Connection is managed at the lowest level
- Client must know what the server can do

- Usually some form of Remote Procedure Call

Point-to-point pattern



- What are the problems with this pattern?
  - Sender must know receiver's location
  - Communicating with many receivers is a logistical headache
    - Cost of update can be high
  - May involve duplication of effort
    - Each node may require data transformation
  - Both must be available at the same time
  - Client must know what the server can do (hardcoded)

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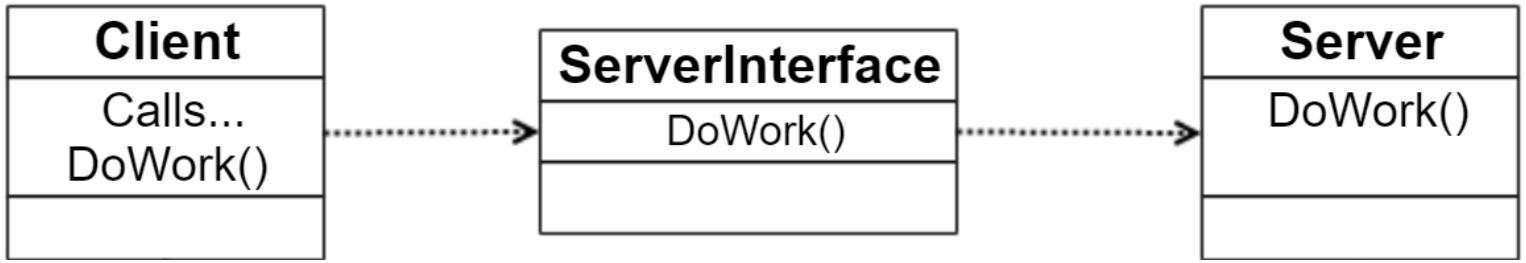
Using an Interface

- The Broker pattern, and many other Service Oriented Architectures (SOAs) use an interface.
- This means that the client can make calls to a known interface
  - The interface advertises functionality but without implementation details
- The client now isn't reliant on what the server itself does under the hood
  - So the implementation of the server can be changed at any time.

- Usually the interface is advertised to the client in some way.
  - This *could* be at runtime, which means no need to hardcode the remote calls.



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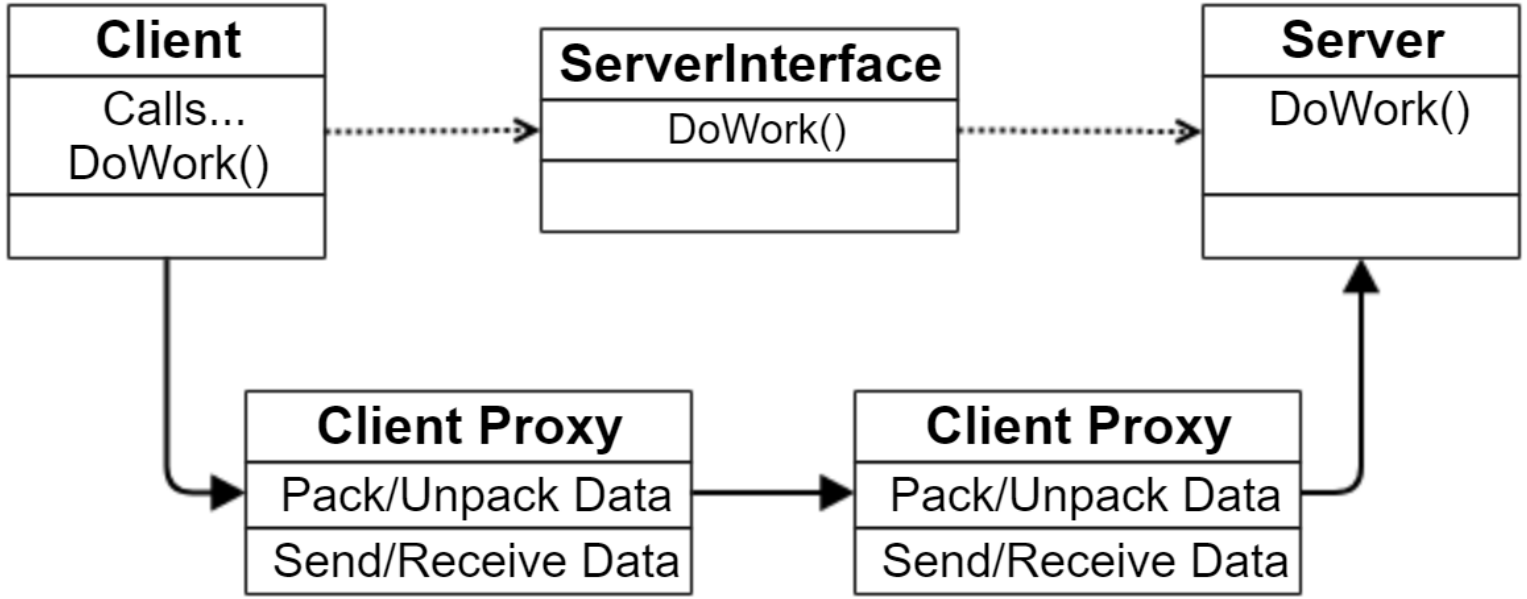
Using a Proxy

- Proxies manage all of the message sending/receiving/packing/unpacking/security/etc.
- Client understands the server interface but REAL data is sent through the proxy
- This helps improve loose coupling – the client no longer needs to manage the lower-level data transmission tasks.



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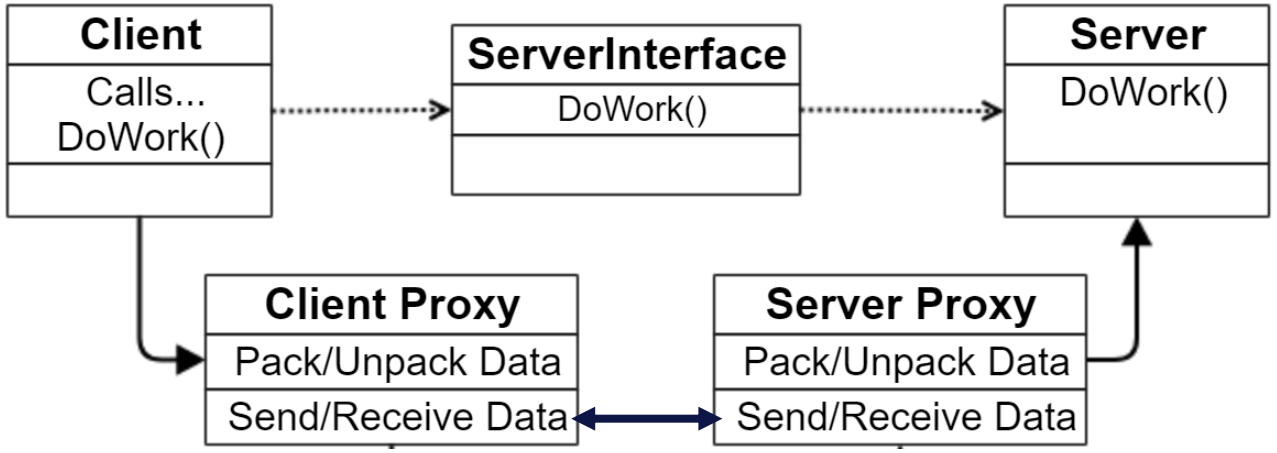
Using a Proxy

- But there is still a problem… What is the problem here?
  - The client still needs to know where the server is before it can call it
  - Application developers should not have to learn about the details of cross-process remote communications.
  - To maintain flexibility, developers must avoid hard-coding the location of specific components



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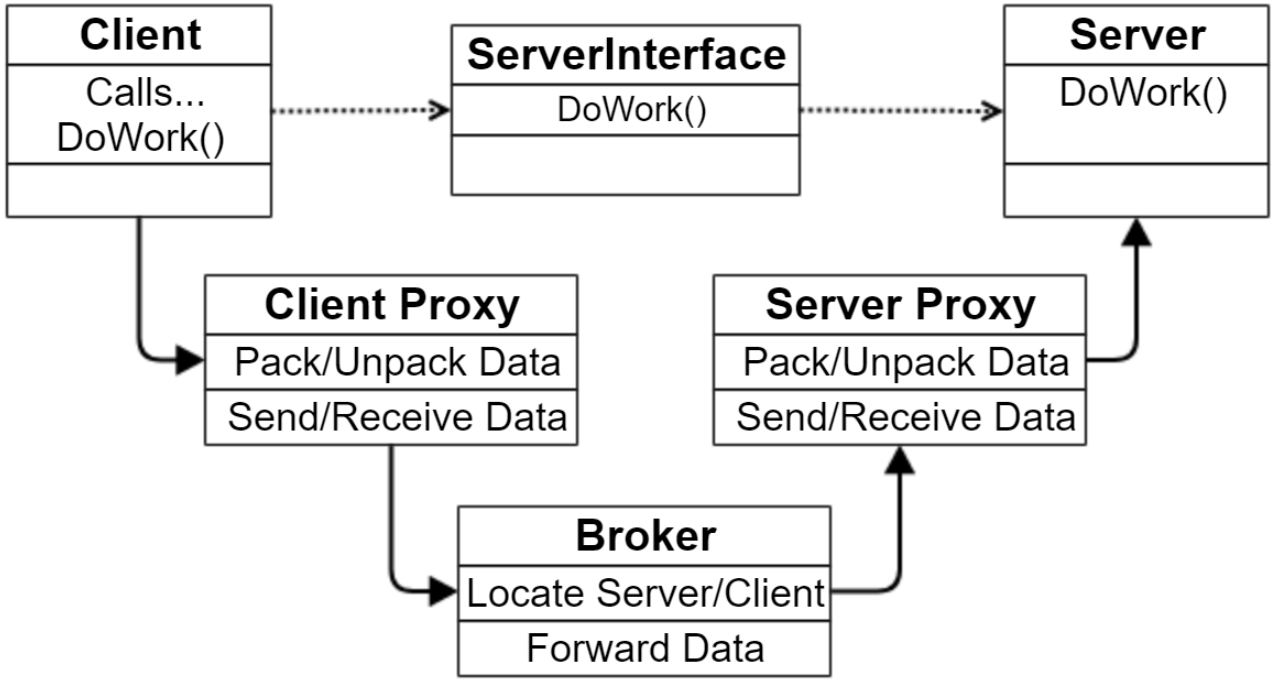
Broker

- The Broker manages the location of the client/server and acts as a messenger between the two.
- To the Client and the Server, calls are now routed properly without having to know and store locations.



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Broker Pattern

- Hides the implementation details of remote service invocation by encapsulating them into a layer other than the business component itself
- Provides an interface to the client that allows it to invoke methods just as it would invoke any local interface
- Methods inside the client interface trigger services to be performed on remote objects…

  …or remote services
- Broker may determine location of services…

  …client doesn't need to know

  …developer doesn't need to hard-code

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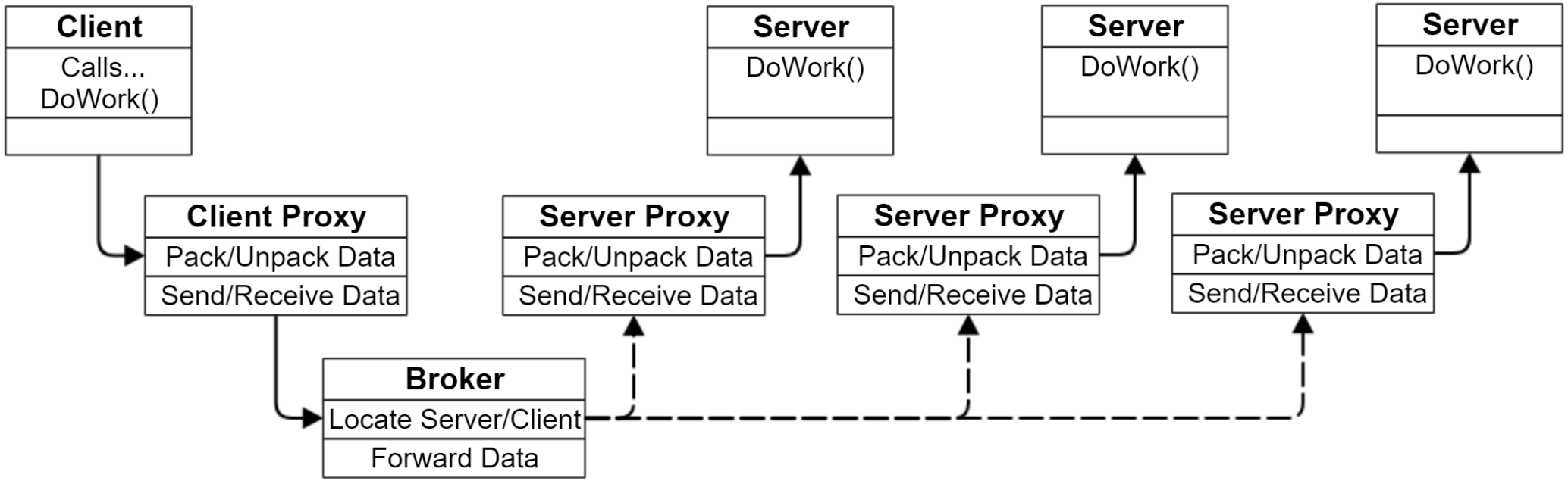
Broker pattern

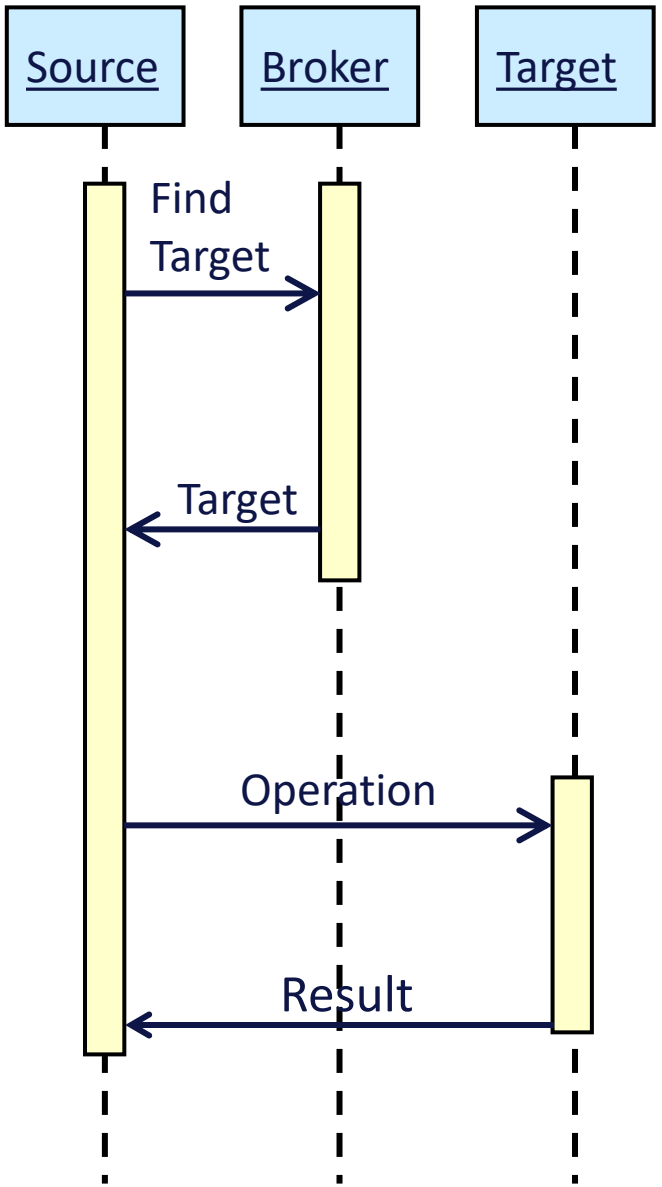
- Required services may be anywhere (and could be multiple different servers)
- Broker acts as intermediary
- Broker maintains a directory
- Client uses service interface
- Broker provides details to proxies for connection
- Servers inform broker of their existence



Direct broker

- Broker establishes communication between endpoints
- Once communication is established, system is point-to-point
- Addresses and connections are centrally managed

- Used by .NET remoting
- Example: naming services
  - Active Directory
  - DNS

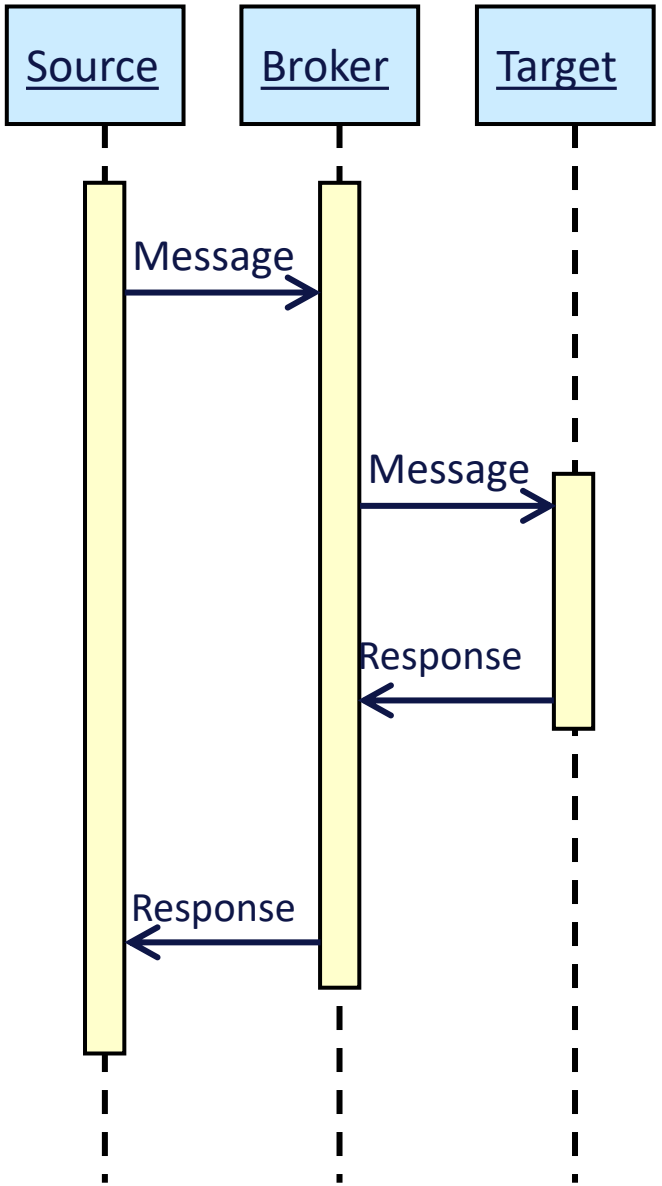


Indirect broker

- All communication is with broker
- Broker passes messages to recipients
- Addresses and connections are centrally managed

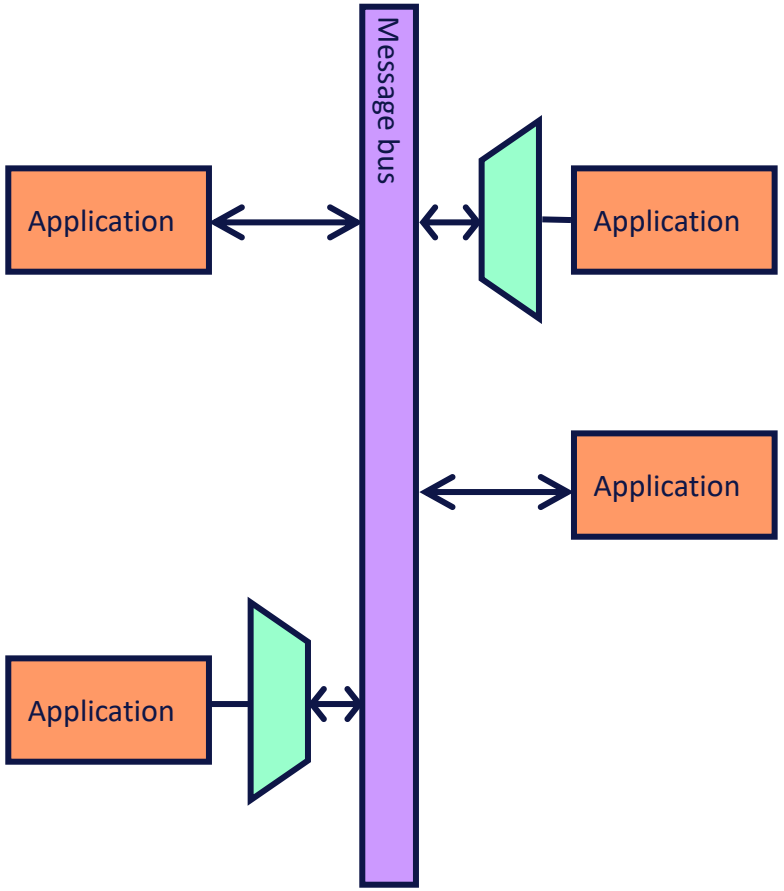
- Not as efficient as direct broker
- Single point of failure
- Single bandwidth bottleneck

- **Q:** In what situation is an indirect broker useful?
  - Brokers can add additional protections, e.g. a firewall



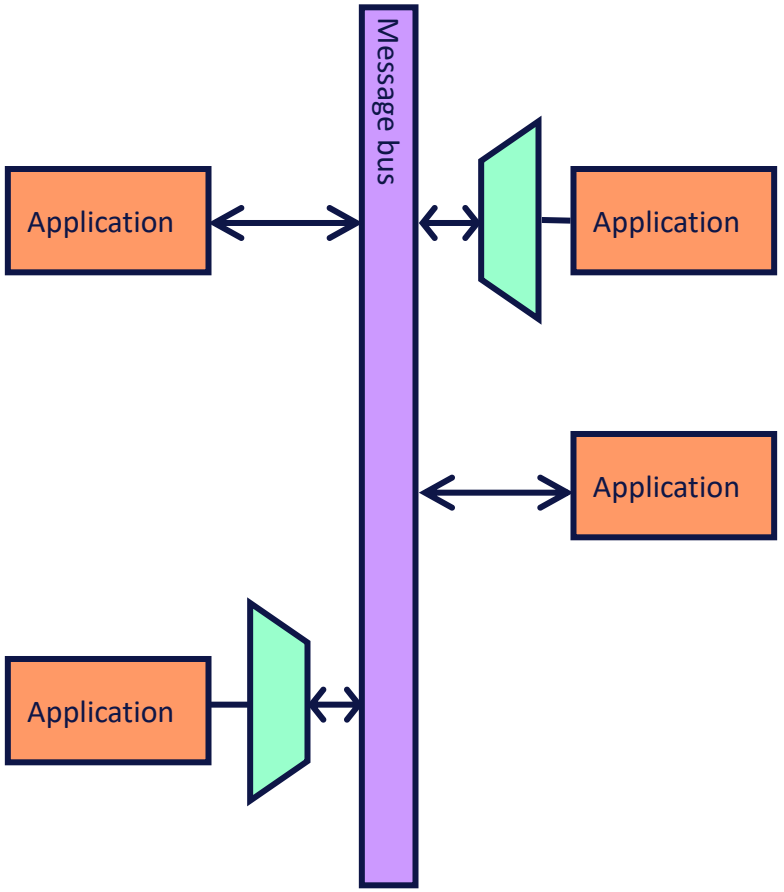
Message Bus

- The *Message bus* pattern defines a standard way of moving data packets from a target system to an intermediary store and transport
- ***May*** provide routing for messages



Message bus

- Applications use message bus middleware to communicate
  - Directly to message bus
  - Indirectly via adapters
- Messages may contain target information
  - May contain routing
  - May be a routing agent in message bus
- Message bus acts as universal connector
- Messages must use a common, standardised, data model
  - Although this could be very basic



Message bus features

- Common communication infrastructure
  - Cross-platform, cross-language
  - Universal adapter for messages
  - May include routing facilities
    - Message may contain info on who it's for (message bus can do routing if required)
    - Or endpoints may keep track of what is for them (endpoints listen for messages on unrouted bus)
- Adapters
  - Each (application/system/protocol) needs a specific message adapter
  - Can be event generators or data transforms
  - Many filtering operations can be done on messages
- Common command structure
  - Message bus can carry instructions as well as data

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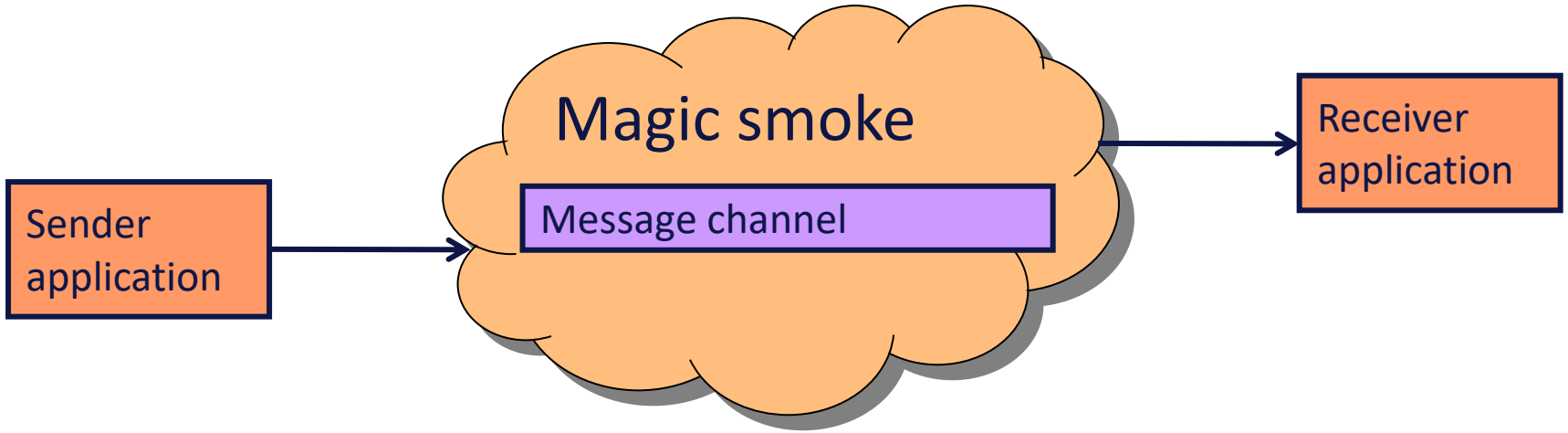
Message channels

- How is Message Bus implemented?



Applications communicate by use of 'Message Channels' which are implemented in "some way"

- Message channel carries specific message to specific destinations OR is common
- Applications may have multiple channels
- Sender puts message on specific channel – receiver retrieves from specific channel
  - Something like streams or pipes

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Message Bus Queueing

- Generally gives us the option of putting a message into a queue whilst the recipient is not available.
  - Currently “down”
  - Overloaded
- Also may give options for multiple instances of the same recipient software to take from a queue so as to spread the load across devices
- Individual queues can be created where the message bus uses routing

- There are lots of message bus with queue options, e.g.
  - RabbitMQ (open-source)
  - Amazon Simple Queue Service (Cloud)

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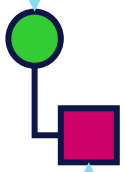
Messages

- Header
  - Contains information about the message
    - Destination
    - Source
    - Type
      - DocumentMessage, CommandMessage…
    - Sequence number
  - May be modified by message handling/routing systems
- Body
  - Payload information
  - Generally transmitted verbatim



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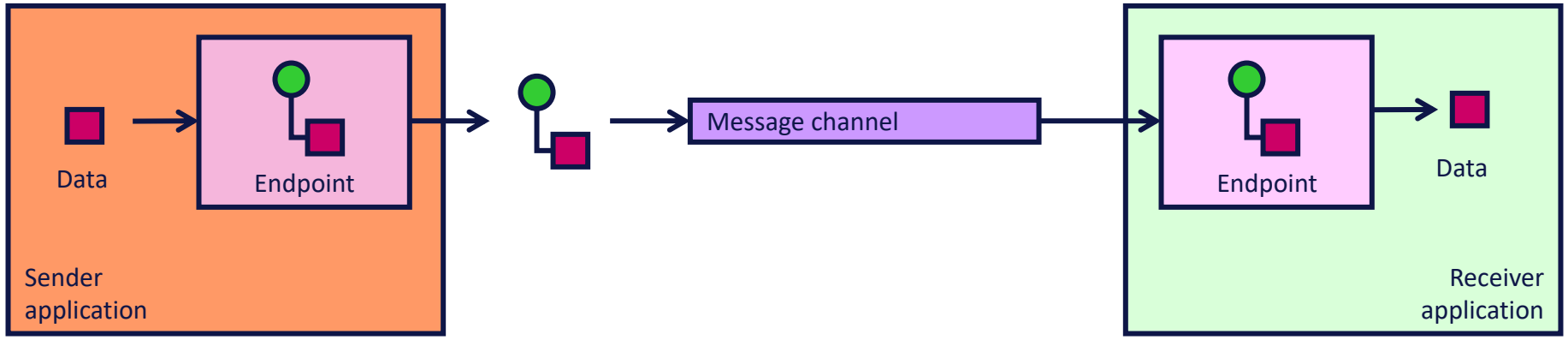
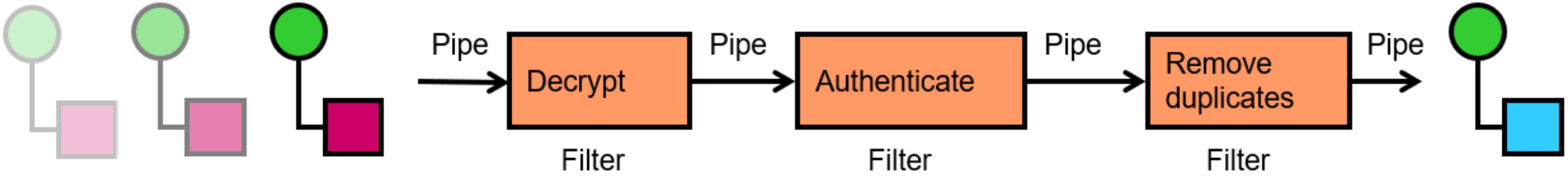
Endpoints

- Sender and receiver are different applications
- May be on different OS or platform
- Endpoint converts data to message, message to data
- Endpoint is channel-specific (so there could be more than one endpoint)
- Manages the lower layers of the protocol so the application itself doesn't need to be concerned with the intricacies of messaging



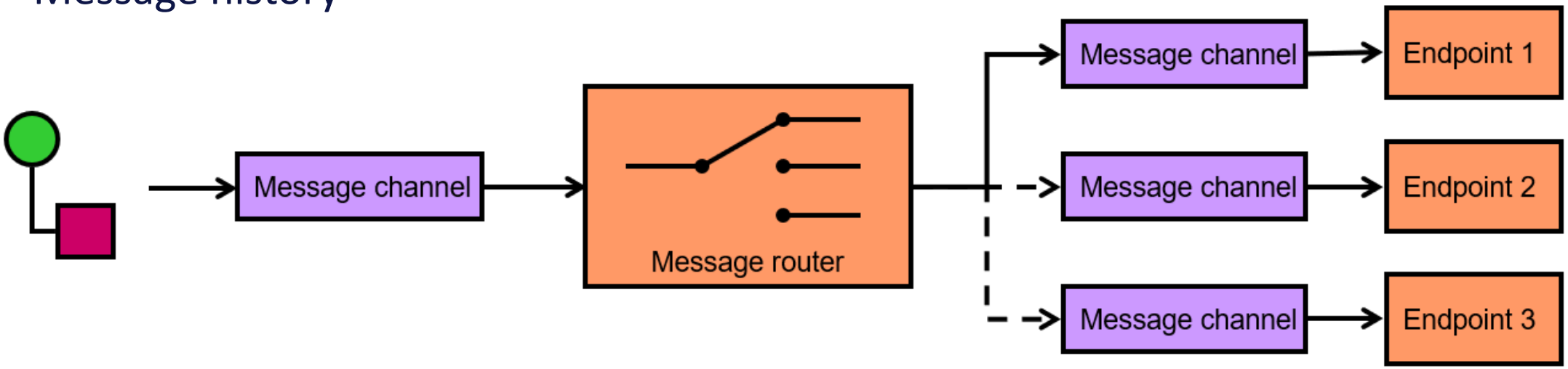
Pipes and filters

- Pipes take messages from one point to another
  - Can be a message channel (if resources permit)
  - In-memory queue may be more efficient if co-located
- Filters (selectively) process messages
  - Performing specific operations
  - May modify message header or body
- Allows common work to be done in the pipe rather than in the application
  - Meaning that the application is only focused on the business logic



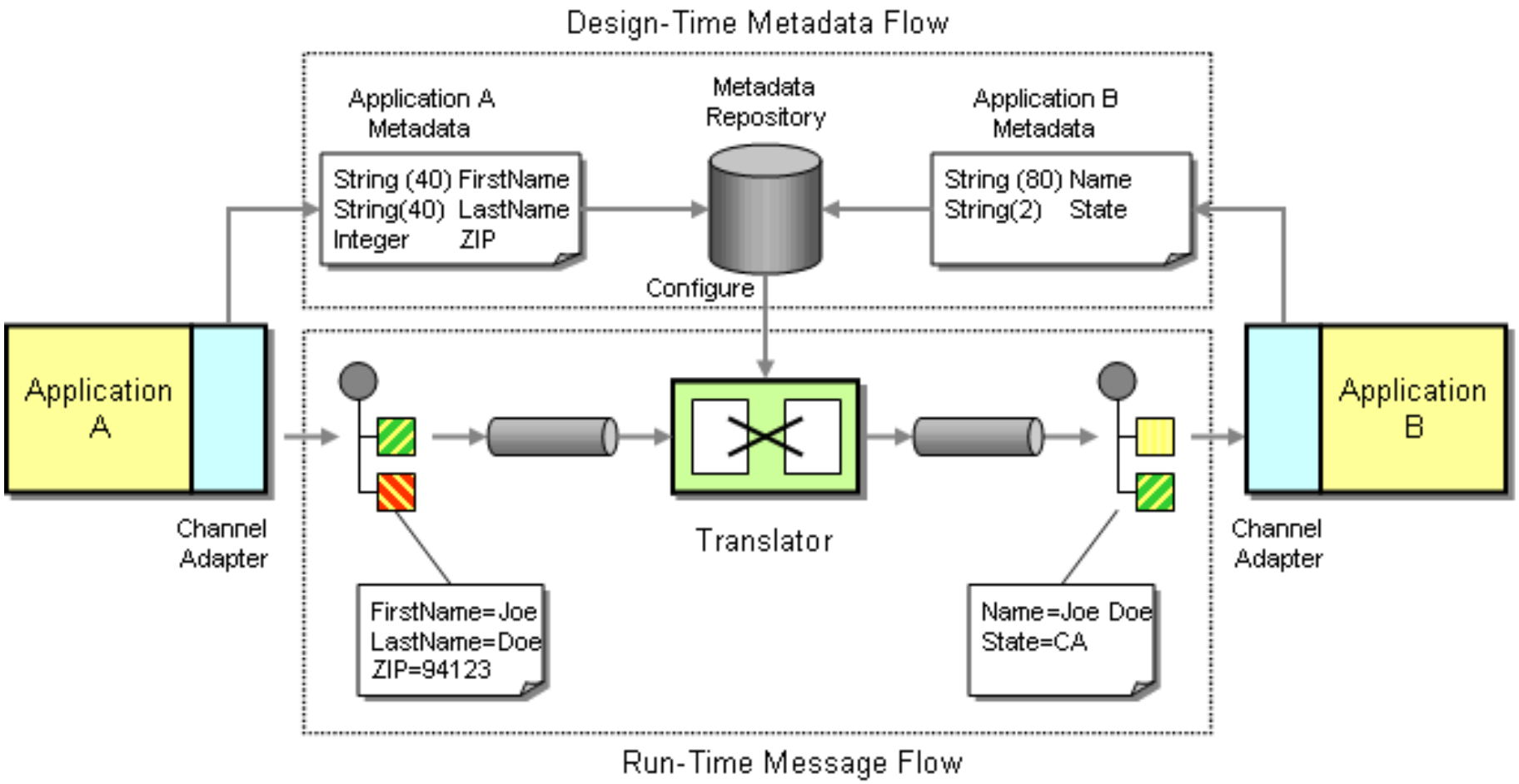
Routing

- Decouple individual processing steps so that messages can be passed to different filters or endpoints depending on a set of conditions
- Might depend on
  - Message content
  - Message type
  - Message header
  - Message history



Transformations

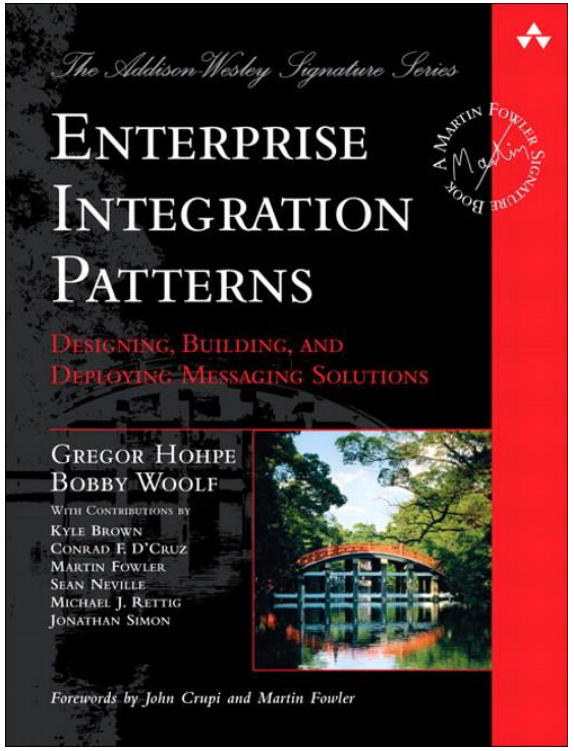
- Filters may transform messages (can be a two-way transformation)
- Change data format and object model based on prior knowledge (metadata)



Book

Enterprise Integration Patterns:
Designing, Building,
and Deploying Messaging Solutions

Gregor Hohpe, Bobby Woolf



Free resources:

https://www.enterpriseintegrationpatterns.com/patterns/messaging/

https://www.enterpriseintegrationpatterns.com/ramblings/eip1_examples_updated.html

https://www.enterpriseintegrationpatterns.com/patterns/conversation/

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Reflection

We've looked at:

- Architectural Pattern Types
- Message-based Patterns
- Messaging
- Pipelines
- Transformations

In the next session we will continue to look at Design Patterns, explicitly Software Design Patterns

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